



How Hybrid Approach Can Enhance Quality Of Ultrasonic Images

Ultrasonic imaging is simple and cost-effective, especially for use in remote places, but it is not as good as CT or MRI imaging. However, with each passing day, and with technology getting better and better, software-based ultrasonic imaging systems are becoming better too. It is just a matter of time before these start replacing CT and MRI, according to the authors

Ayush Dogra and **Sanjeev Virdi** belong to Biomedical Instrumentation Unit, CSIR-CSIO, Chandigarh. **Apoorav Maulik Sharma** is from Department of ECE, UIET, Panjab University, Chandigarh, and **Bhawna Goyal** is from Department of ECE, Chandigarh University, Punjab

Ultrasound imaging has been around for many decades now. Regular enhancements in its features and capabilities, and the rapid miniaturisation of ultrasound devices has helped it thrive in the field of medical diagnosis. It started way back in 1794 when, Lazzaro Spallanzani, an Italian physiologist came to know about the phenomenon of echolocation (using ultrasound waves to hunt and trace path) among bats, which later formed the basis of ultrasonic imaging.

Further, in 1877, piezoelectricity was discovered by Pierre Currie and Jacques Currie. Piezoelectric effect is used in transducers for ultrasonic imaging; it converts electric current to longitudinal acoustic waves. In 1915, the first transducer was designed to detect objects under the sea. In early 1940s, first instance of using sonography for medical diagnosis came into light when a neurologist named Karl Dussik used it for detecting brain tumors.

In late 1940s came the first A-mode ultrasound device for detecting gallstones, and between 1949 and 1951, 2D B-mode ultrasound device was introduced. Later, in 1970s, further developments came in ultrasound devices in the form of Doppler ultrasound and Colour Doppler ultrasound. Introduction of more sophisticated ultrasound devices with 3D capabilities and M-mode visualisation took off in 1990s.

Ultrasonic imaging is done using acoustic waves of ultra-high frequency. The audible range of human ear is 20Hz-20 kHz. For ultrasonic imaging, acoustic frequencies of

about 10MHz are used normally. There are two main reasons for choosing high frequencies for this operation:

1. If normal audible frequencies are used for imaging, it may annoy the patients by creating loud noise, and that too in an environment where it is strictly undesirable.
2. Higher frequencies suffer low losses due to various factors. So images captured using high frequencies will have better axial resolution.

Now, before going into the working, we must understand certain aspects of ultrasonic imaging. It is normally done in four modes, which are:

1. *A-mode*. In this mode a single transducer is used to project, detect and plot the wave with respect to the depth of the target. Diagnosis of a specific tumor is also done using therapeutic A-mode ultrasonic imaging.

2. *B-mode*. This is the type through which the echoed signal can be viewed as a 2D image on the screen. To achieve this a linear array of transducers is used to simultaneously scan through the targets by moving or swinging the transducer.

3. *M-mode*. Known as Motion mode, in this, 2D images or B-mode scans of an organ are run in a sequence according to transducer position on a screen, so that information from multiple angles could be obtained to make more accurate conclusions.

4. *Doppler mode*. This mode is generally used to visualise the body fluids that are in motion (usually blood). The Doppler Effect comes into picture as the moving body fluids change frequency of the echoes sent back.